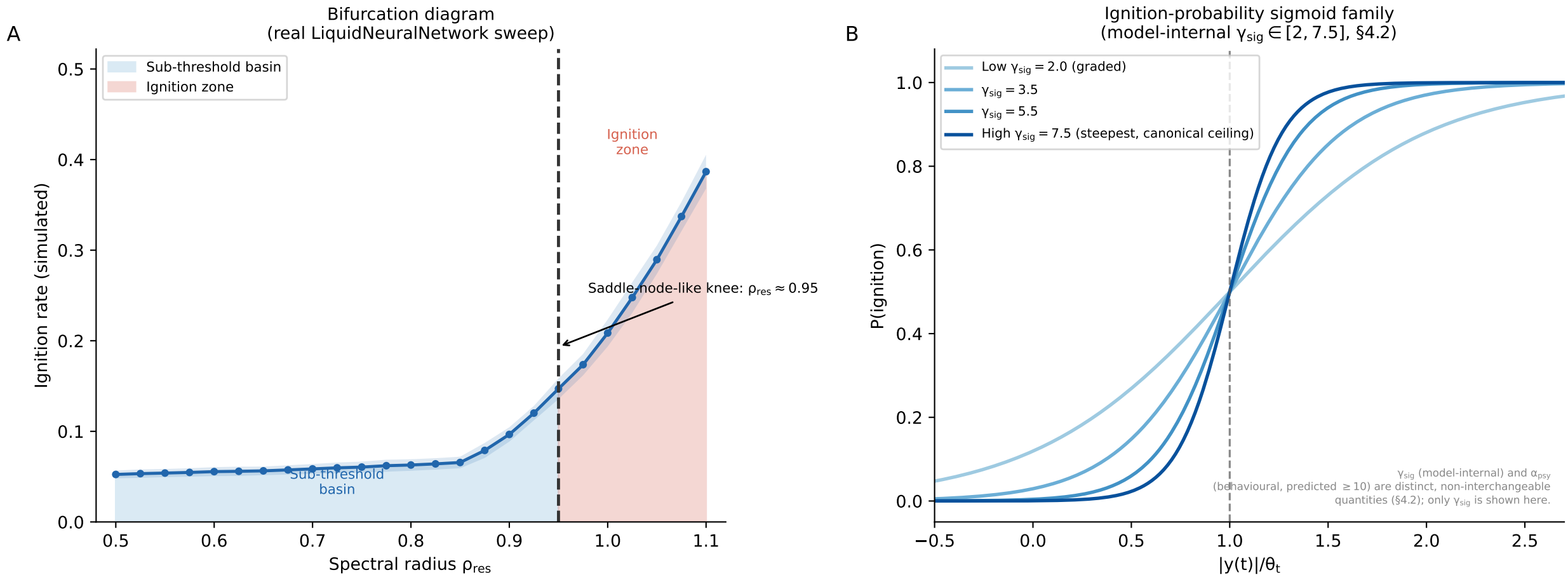


Figure 3 — APGI-LNN Bifurcation Analysis: Parameter Sweep and Ignition Manifold
(Paper 2, §4.5)



Heterogeneous- τ form used (§4.1). Panel A: real LiquidNeuralNetwork simulation, spectral radius ρ_{res} swept over $[0.5, 1.1]$, $N=1200$ steps $\times$ 12 seeds per point; S_t proxy = $||x(t)||$ (reservoir-state norm) vs. $\theta_t = 2.4$, post-ignition reset $x \leftarrow \rho_S \cdot x$ ($\rho_S = 0.5$). Proof-of-concept scale only; biologically realistic validation ($N \geq 1,000$ units) pending.